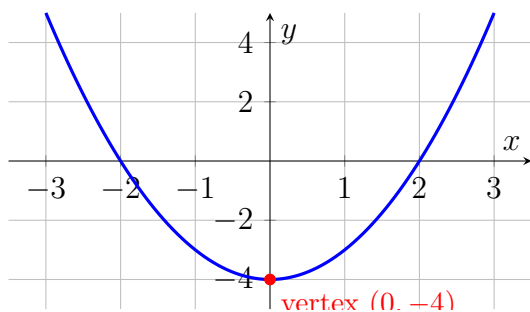


Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____

Quadratic Functions: Vertex, Max & Min

Example. For $y = x^2 - 4$ the vertex is $(0, -4)$, which is a **minimum**. The graph *decreases* for $x < 0$ and *increases* for $x > 0$.



Directions: for each function, state max or min, give the vertex, and the increasing / decreasing intervals.

1. $y = x^2 + 2$

3. $y = (x - 3)^2$

2. $y = -x^2 + 5$

4. $y = -(x + 1)^2 - 2$

Challenge (same sheet).

5. $y = (x - 2)^2 - 9$ — give the vertex, both intervals, *and* the two x -intercepts.

Answer Key

1. Min $(0, 2)$; decreasing $x < 0$, increasing $x > 0$
2. Max $(0, 5)$; increasing $x < 0$, decreasing $x > 0$
3. Min $(3, 0)$; decreasing $x < 3$, increasing $x > 3$
4. Max $(-1, -2)$; increasing $x < -1$, decreasing $x > -1$
5. Min $(2, -9)$; decreasing $x < 2$, increasing $x > 2$; x -intercepts $x = -1, 5$